
Technology transfer package — Toscanini invention

invr452-fresh-evaporative-vaccine-coolerae7a00faa34a · run run · maturity EXPERIMENT_READY · 3D class ENGINEERING_3D

Problem definition

- Problem Id: r452-fresh-evaporative-vaccine-cooler
- Device: passive evaporative-cooled vaccine storage vessel for last-mile clinics
- Failure: vaccine doses spoil in transit because the passive evaporative cooler loses cooling margin within hours when ambient humidity rises, the water-wick channel clogs with dissolved salts, and clinic staff have no indication of how much cooling margin remains before the interior exceeds 8 C
- Failure Mode: cooling_margin_loss
- Constraint: cooler must hold a...

Failure mode being addressed

6 failure modes recorded. Primary: cooling_margin_loss — trigger: the operating condition stated in the problem (mechanism detail beyond the problem statement and trigger quantification NOT ESTABLISHED)

Mechanism

Toscanini investigated the problem “passive evaporative-cooled vaccine storage vessel for last-mile clinics” and, after 2 architecture generations of challenge-and-evolution, its current invention is generation 2. The causal mechanism is A composite radiative coating embedded with microencapsulated phase-change materials that alter their infrared emissivity based on temperature thresholds applied at passive evaporative-cooled vaccine storage vessel for last-mile clinics. The expected measurable effect is Dynamic temperature regulation within the 2-8°C range with 40% improved efficiency during temperature spikes without external energy input. The invention's maturity is recorded honestly: thi...

Causal delta

What distinguishes this generation from its parent is the explicit causal change recorded in the evolution: A composite radiative coating embedded with microencapsulated phase-change materials that alter their infrared emissivity based on temperature thresholds — the architecture delta versus the parent generation is recorded in the run state and inspectable in the package.

Architecture

- heat source characterization — status ENGINEERING_PROPOSED
- conduction path / spreader — status ENGINEERING_PROPOSED
- reject interface — status ENGINEERING_PROPOSED
- safety limit interlock — status ENGINEERING_PROPOSED

Subsystem breakdown

- heat source characterization — status ENGINEERING_PROPOSED: NOT ESTABLISHED (requires design work)
- conduction path / spreader — status ENGINEERING_PROPOSED: NOT ESTABLISHED (requires design work)

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- reject interface — status ENGINEERING_PROPOSED: NOT ESTABLISHED (requires design work)
 - safety limit interlock — status ENGINEERING_PROPOSED: NOT ESTABLISHED (requires design work)

Governing equations

- TH-002: $q = h * A * dT$ — Convective heat transfer (Newton's law of cooling)
 - TH-003: $R_{total} = \sum(L_i / (k_i * A_i))$ — Series thermal resistance
 - TH-001: $q = -k * A * dT/dx$ — Fourier conduction
- (3 total; full bindings in EQUATION_REGISTRY.json)

Parameter definitions

- CP-001 heat generation at source [q W] = UNKNOWN (no sourced value) (UNKNOWN)
- CP-002 spreader conductivity [k_i W/(m.K)] = UNKNOWN (no sourced value) (UNKNOWN)
- CP-003 interface contact resistance [R_{total} K/W] = UNKNOWN (no sourced value) (UNKNOWN)
- CP-004 reject-path area [A m²] = UNKNOWN (no sourced value) (UNKNOWN)
- CP-005 allowable temperature rise [dT K] = UNKNOWN (no sourced value) (UNKNOWN)
- CP-006 ambient worst case [NOT_ASSIGNED UNKNOWN] = UNKNOWN (no sourced value) (UNKNOWN)

Constraints

- Stated Constraint: cooler must hold a 2-8 C interior for at least 48 hours in 40 C ambient, fit a 300 mm length by 250 mm width footprint with 400 mm height, carry a 5 L vaccine payload, use no electricity, and signal the remaining cooling margin without any power source
- Failure Mode Of Device: cooling_margin_loss

Engineering assumptions

- Assumption: mechanism transfers across domain context
- Basis: MODELLED
- Test: Compare thermal performance of coated vs uncooled units under controlled temperature cycling from 15°C to 45°C ambient
- Assumption: searched sources approximate the prior-art universe
- Basis: MODELLED
- Test: expand sources (see next_best_action)

Failure modes

- FM-001: cooling_margin_loss — severity UNKNOWN (no sourced severity basis)
- FM-DOM-001: hotspot exceeding material/safety limit — severity UNKNOWN (no sourced severity basis)
- FM-DOM-002: interface delamination raising resistance over life — severity UNKNOWN (no sourced severity basis)
- FM-DOM-003: coolant/flow path blockage (if active) — severity UNKNOWN (no sourced severity basis)
- FM-DOM-004: thermal runaway coupling to electronics — severity UNKNOWN (no sourced severity basis)
- FM-DOM-005: condensation at cold surfaces — severity UNKNOWN (no sourced severity basis)

Verification plan

- VF-001: result NOT_TESTED — Compare thermal performance of coated vs uncooled units under controlled temperature cycling from 15°C to 45°C ambient
- VF-002: result NOT_TESTED — run killer experiment: falsification_test_from_candidate (EIG/cost 0.4557178723356231)
- VF-003: result NOT_TESTED — temperature-rise mapping at max duty
- VF-004: result NOT_TESTED — temperature-rise mapping at maximum duty with IR or embedded sensing (measured quantity: temperature rise)
- VF-005: result NOT_TESTED — thermal-resistance trend over accelerated cycling (measured quantity: mechanical stress / fatigue endurance)
- VF-006: result NOT_TESTED — temperature-rise mapping at max duty
- VF-007: result NOT_TESTED — worst-case ambient soak with staged power steps and stability monitoring (measured quantity: power / energy deposition)

Build / manufacturing pathway

- WP-01: test article first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed); measurement physical bench/clinical experiment — NOT simulation
- WP-02: test article constraint-compliance envelope article; measurement constraint-compliance characterization
- WP-03: test article article for VF-001; measurement Compare thermal performance of coated vs uncooled units under controlled temperature cycling from 15°C to 45°C ambient
- WP-04: test article article for VF-002; measurement run killer experiment: falsification_test_from_candidate (EIG/cost 0.4557178723356231)
- WP-05: test article article for VF-003; measurement temperature-rise mapping at max duty
- WP-06: test article article for VF-004; measurement temperature-rise mapping at maximum duty with IR or embedded sensing (measured quantity: temperature rise)
- WP-07: test article article for VF-005; measurement thermal-resistance trend over accelerated cycling (measured quantity: mechanical stress / fatigue endurance)
- WP-08: test article article for VF-006; measurement temperature-rise mapping at max duty
- WP-09: test article article for VF-007; measurement worst-case ambient soak with staged power steps and stability monitoring (measured quantity: power / energy deposition)

Regulatory posture

problem context (canonical applicability): UNKNOWN. Context requirement [UNKNOWN]: UNKNOWN — regulatory applicability not determinable from the canonical problem text. regulatory pathway: NOT ESTABLISHED (no determination exists for this invention; the pathway is never inferred from device class alone — the buyer's first regulatory action is the determination itself)

Decisive experiment

- Selected: falsification_test_from_candidate
- Eig: 0.4557178723356231
- Eig Per Cost: 0.4557178723356231
- Definition: physical bench/clinical experiment — NOT simulation
- Hypotheses: • H_effect_holds
- H_effect_fails
- Options Ranked: • falsification_test_from_candidate
- contradiction_attacking_experiment

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- Epistemic Note: all priors MODEL_DERIVED; no EXPERIMENTALLY_ESTIMATED inputs exist at this stage (Art. XXVIII/XXXVIII: no synthetic shortcut to physical observation)

Expected measurements

- Compare thermal performance of coated vs uncooled units under controlled temperature cycling from 15°C to 45°C ambient
 - run killer experiment: falsification_test_from_candidate (EIG/cost 0.4557178723356231)
 - temperature-rise mapping at max duty
 - temperature-rise mapping at maximum duty with IR or embedded sensing (measured quantity: temperature rise)
 - thermal-resistance trend over accelerated cycling (measured quantity: mechanical stress / fatigue endurance)
 - temperature-rise mapping at max duty
- (all NOT_TESTED — expected values are UNKNOWN until the experiment runs)

Acceptance criteria

Pre-registered rule (from VF-001): pre-registered pass/fail against the problem-stated threshold(s) (SOURCE_FACT, problem statement; the sourced values and their provenance are recorded on the corresponding design-input records)

Unresolved technical risks

The run records its unknowns in the package's engineering definition; each unknown is material and none is suppressed. Physical output is NOT ESTABLISHED — no measurement of this invention exists.

3D explanation

The parametric engineering model (CadQuery/OCCT authority) with STEP/STL exports, measured key dimensions and validation gates; the derivation chain is in MODEL/ENGINEERING_PROVENANCE.json and the independent checks in MODEL/3D_EVIDENCE/. Renders are presentation artifacts — never physical truth.

Artifact references

every layer of this package is named in PACKAGE_MANIFEST.json with per-file sha256; the machine-readable layers are ENGINEERING_TRACEABILITY, EQUATION_REGISTRY, UNKNOWN_ROADMAP, VALIDATION_ECONOMICS, MATURITY_BASIS, LOOP_STATE, PROVENANCE (top level) and the MODEL/ tree

Epistemic boundary: DESIGNED / SIMULATED / EVIDENCE-SUPPORTED / EXPERIMENTALLY VERIFIED states come only from the canonical invention object. EXPERIMENTALLY VERIFIED requires real reality-loop evidence. Potential IP territory — formal legal review required. Toscanini performs technology discovery; patentability, claim drafting, infringement/FTO and legal opinions are matters for qualified IP counsel.